

- 2** A program reads string data from a text file into a linear queue and then compresses the data.

The queue is created using the global 1D array `Queue`. The queue can store up to 100 elements. Each element is initialised to the empty string ""

The queue has two pointers that are global to the program:

- `QueueHead` that points to the index of the first element in the queue, initialised to -1
- `QueueTail` that points to the index of the last element in the queue, initialised to -1

The global variable `NumberItems` stores the number of elements stored in the queue, initialised to 0

- (a)** Write program code to declare and initialise `Queue`, `QueueHead`, `QueueTail` and `NumberItems`

Save your program as **Question2_N25**.

Copy and paste the program code into part **2(a)** in the evidence document.

[2]

- (b)** The function `Enqueue()` takes a string as a parameter. The function checks if the queue is full, and returns Boolean `FALSE` if the queue is full.

If the queue is not full, the parameter is inserted into the next position in `Queue`. The function updates the appropriate pointer(s), updates `NumberItems` and then returns Boolean `TRUE`

Write program code for `Enqueue()`

Save your program.

Copy and paste the program code into part **2(b)** in the evidence document.

[5]

- (c)** The function `Dequeue()` returns the string "False" if the queue is empty.

If the queue is not empty, the function returns the next element in the queue. The function updates the appropriate pointer(s) and updates `NumberItems`

Write program code for `Dequeue()`

Save your program.

Copy and paste the program code into part **2(c)** in the evidence document.

[3]